

Correction - Distributivité simple

Exercice 1

a. $3(x + 4)$

$$= 3 \times x + 3 \times 4$$

$$= 3x + 12$$

b. $5(y - 2)$

$$= 5 \times y - 5 \times 2$$

$$= 5y - 10$$

c. $7(2x + 1)$

$$= 7 \times 2x + 7 \times 1$$

$$= 14x + 7$$

d. $-2(x + 6)$

$$= -2 \times x + (-2) \times 6$$

$$= -2x - 12$$

e. $-4(3 - a)$

$$= -4 \times 3 - (-4) \times a$$

$$= -12 + 4a$$

f. $x(x + 5)$

$$= x \times x + x \times 5$$

$$= x^2 + 5x$$

Exercice 2

$$A = 2(x + 3) + 5x$$

$$A = 2 \times x + 2 \times 3 + 5x$$

$$A = 2x + 6 + 5x$$

$$A = 7x + 6$$

$$B = 4(2x - 1) - 3$$

$$B = 4 \times 2x - 4 \times 1 - 3$$

$$B = 8x - 4 - 3$$

$$B = 8x - 7$$

$$C = 3(x + 2) + 2(x - 5)$$

$$C = 3 \times x + 3 \times 2 + 2 \times x - 2 \times 5$$

$$C = 3x + 6 + 2x - 10$$

$$C = 5x - 4$$

$$D = 5(2y + 1) - 3(y + 4)$$

$$D = 5 \times 2y + 5 \times 1 - 3 \times y - 3 \times 4$$

$$D = 10y + 5 - 3y - 12$$

$$D = 7y - 7$$

$$E = -2(3x - 4) + 6x$$

$$E = -2 \times 3x - (-2) \times 4 + 6x$$

$$E = -6x + 8 + 6x$$

$$E = 8$$

$$F = x(x + 2) + 3(x - 1)$$

$$F = x \times x + x \times 2 + 3 \times x - 3 \times 1$$

$$F = x^2 + 2x + 3x - 3$$

$$F = x^2 + 5x - 3$$

Exercice 3

$$G = 5 + (3x - 2)$$

$$G = 5 + 3x - 2$$

$$G = 3x + 3$$

$$H = 7x - (2x + 4)$$

$$H = 7x - 2x - 4$$

$$H = 5x - 4$$

$$I = 4 - (x - 6) + 3x$$

$$I = 4 - x + 6 + 3x$$

$$I = 2x + 10$$

Exercice 4

1. Aire = $3(x + 5)$

2. $3(x + 5)$

$$= 3 \times x + 3 \times 5$$

$$= 3x + 15$$

3. Pour $x = 2$: $3 \times 2 + 15 = 6 + 15 = 21$

L'aire du rectangle est 21 cm^2 .